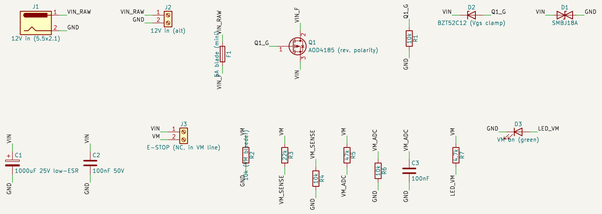
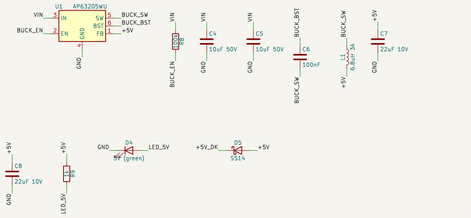
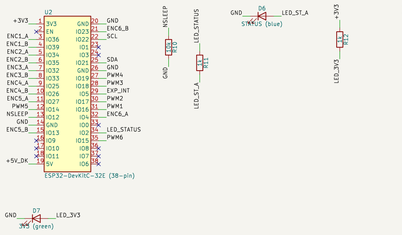
POWER IN, REVERSE PROTECTION, E-STOP (no relay: NC contacts in VM line)



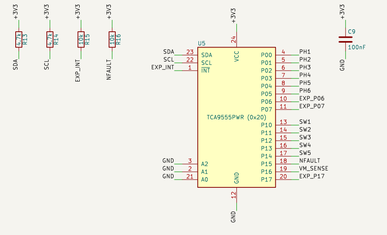
5 V BUCK + DEVKIT 5V FEED



ESP32-DevKitC-32E (plug-in, 2x 1x19 female headers)

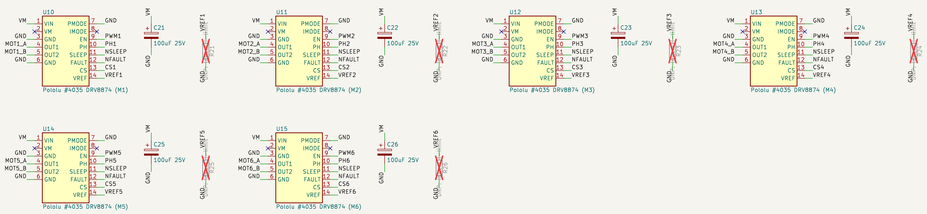


I2C: TCA9555 (0x20) + ADS7828 (0x48)

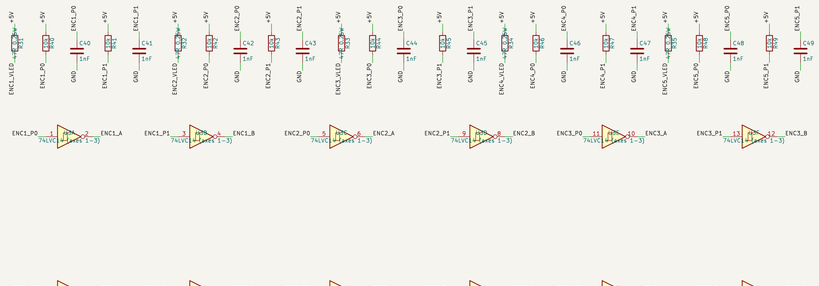




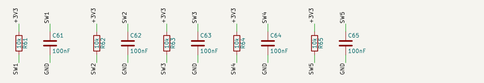
MOTOR DRIVERS: 6x Pololu DRV8874 carrier (plug-in) — PIN ORDER PLACEHOLDER



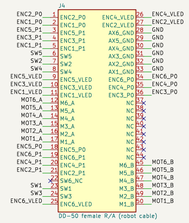
ENCODER FRONT END: 47R VLED, 10k pull-ups to 5V, 1nF, 74LVC14 at 3.3V



HOME SWITCHES (to TCA9555 P1.0–P1.4)



ROBOT CABLE: DD-50 female (SCORBOT-ER 4u manual Rev E)



POWER FLAGS



MOUNTING



NOTES

1. Pololu DRV8874 carrier pin ORDER is a placeholder (names are correct). Fix design.CARRIER_ROWS and regenerate.
2. DB50 pinout from SCORBOT-ER 4u User Manual ch. 8 (same as ER III Table D-1). Pins 37–44 unused, 22 = gripper switch (no connection in arm).
3. 74HC14 symbol is used for the 74LVC14 (same pinout). Powered from +3V3: inputs are 5 V tolerant.
4. Carrier current limit with SLEEP at 3.3 V is about 2.9 A (VREF pulled to SLEEP by 10k on the carrier, CS has 2.49k). R21–R26 (DNP) lower it.
5. ADS7828 uses its internal 2.5 V reference: CS reads 1.12 V/A, so readings clip above about 2.2 A.

Pololu carrier pin ORDER is a placeholder — see README
Generated by generator/gen_schematic.py from design.py

Sheet: /
File: scorbot_controller.kicad_sch

Title: Scorbot ER-4u ESP32 master board

Size: A1	Date: 2026-10-09	Rev: 0.1 (generated)
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1